integration by Parts

- integration by Parts is used when you cannot Solve the integral by using regular integration or by substitution

Formula: $\int u\,dv = uv - \int v\,du$

u: dv:

you take the derivative of "U" and the integral of "dv" once you have done that Just Follow and Plug in the Parts into the highlighted equation above

"U" can be many things but here is how you can choose a good "u" its called "LiPet" which stands for Logrithims, inverse trig, Polynomials, exPonential, trignometric these are all in order once you have already Picked "u" whatever the left over term will be "dv"

☆ You Sometimes will do More then one integral in a Problem that is Completely Fine

Practice Problems

1) $\int x \cdot LN(x)\, dx$

$u: LN(x)$ $\quad dV: x$

$du = \frac{1}{x}$ $\quad V = \int x \approx \frac{x^2}{2}$

$$uV - \int v\,du$$

$$LN(x) \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{1}{x}$$

$$LN(x) \cdot \frac{x^2}{2} - \frac{1}{2} \int x^{\cancel{2}} \cdot \frac{1}{\cancel{x}}$$

$$LN(x) \cdot \frac{x^2}{2} - \frac{1}{2} \left[\frac{x^2}{2} \right]$$

$$= \boxed{LN(x) \cdot \frac{x^2}{2} - \frac{x^2}{4} + C}$$

- LN(x) was chosen to be our "u" because in "LIPet" L is the first letter which stands for logrithim

2) $\int x^2 e^x\, dx$

$u: x^2$ $\quad dV: e^x$

#1 $du = 2x$ $\quad V = \int e^x \approx e^x$

$$uV - \int v\,du$$

$$x^2 e^x - \int e^x \cdot 2x\, dx$$

- to solve this integral we need to apply another intgration by parts we cannot apply u-sub

$u: 2x$ $\quad dV: e^x$

$du = 2$ $\quad V = \int e^x \approx e^x$

$$uV - \int v\,du$$

$$x^2 e^x - \left[2x e^x - \int e^x \cdot 2 \right]$$

$$x^2 e^x - \left[2x e^x - 2e^x \right]$$

$$\boxed{x^2 e^x - 2x e^x + 2e^x + C}$$

#1 this was chosen to be "u" because because it is a polynomial and from "LIPet" polynomial "P" was the third best option there were no No logrithims or inverse trig but there was a polynomiol

3) $\int \tan^{-1}(x)\,dx$

$u: \tan^{-1}(x) \qquad dv: dx$ #1

$du: \frac{1}{1+x^2} \qquad v = \int dx \approx x$

$uv - \int v\,du$

$\tan^{-1}(x)\cdot x - \int x \cdot \frac{1}{1+x^2}\,dx$

$u = 1+x^2$
$du = 2x \cdot dx$
$\frac{du}{2x} = dx$

$\tan^{-1}(x)\cdot x - \int \cancel{x}\cdot \frac{1}{u} \cdot \frac{du}{2\cancel{x}}$

$\tan^{-1}(x)\cdot x - \frac{1}{2}\int \frac{1}{u}\cdot du$

$$\boxed{\tan^{-1}(x)\cdot x - \frac{1}{2}\cdot \mathrm{LN}|1+x^2| + C}$$

#1 the reason why you can assign $dv = dx$ because you already called $u = \tan^{-1}(x)$ the only thing that leaves you with is dx so you assign $dv = dx$

4) $\int e^x \cos(x)\,dx$

$u = e^x \qquad dv = \cos(x)$

$du = e^x \qquad v = \int \cos(x) \approx \sin(x)$

$uv - \int v\,du$

$e^x \sin(x) - \int \sin(x)\, e^x$

- Solving this integral requires integration by parts

$u = e^x \qquad dv = \sin(x)$

$du = e^x \qquad v = \int \sin(x) = -\cos(x)$

$uv - \int v\,du$

$e^x \sin(x) - \left[-e^x\cos(x) - \int -\cos(x) e^x\right]$

$e^x \sin(x) + e^x\cos(x) - \int \cos(x)\, e^x$

#1

$\int e^x\cos(x) = e^x\sin(x) + e^x\cos(x) - \int \cos(x)\, e^x$

$\int e^x\cos(x) + \int \cos(x)\, e^x = e^x\sin(x) + e^x\cos(x)$

$2\int e^x\cos(x) = e^x\sin(x) + e^x\cos(x))$

$\int e^x\cos(x) = \frac{1}{2}(e^x\sin(x) + e^x\cos(x))$

#1 this was treated as if it was a equation we know that its equal to $\int e^x\cos(x)$ so we combined as if they are like terms

also remember e^x is a exponent which comes before trig "LiPet" thats why it was chosen as "u"

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